2026

COVID Drivers

EVALUATION AND DEPLOYMENT REPORT

LISA OVER

Table of Contents

Introduction ***3***
Objective 1 3
Objective 2 4
Objective 3 4

Section 0: Executive Summary Presentation ***4***

Section 1: Organizational Interpretation of Results ***5***

1.1 Translating Technical Results to Organizational Impact **5**
Performance interpretation 5
Expected organizational outcomes 7
Confidence levels 8
Value proposition 10

1.2 Insights and Recommendations **10**
Key findings 10
Surprising discoveries 12
Strategic recommendations 13
Feature importance interpretation 14
Limitations and caveats 15

1.2 Risk Assessment and Mitigation Strategies **15**
Model risks 15
Data risks 15
Organizational risks 15
Ethical considerations 16
Mitigation approaches 16

Section 2: Project Review and Reflection ***16***

2.1 Project Retrospective **16**
What went well 16
Challenges faced 16
Alternative approaches 16
CRISP-DM reflection 17
Time allocation 17

2.2 Lessons Learned and Future Applications **17**
Technical lessons 17
Process insights 17
Domain knowledge 17
Tool proficiency 17
Future applications 17

Conclusion ***18***

AI Statement of Use ***19***

Bibliography ***19***

Introduction

This project explored a current topic of interest involving Pennsylvania (PA) drivers and driving behaviors that may have changed during and after the COVID-19 pandemic. A majority of Americans believe drivers have been more aggressive and distracted since the lockdown. Several studies were reviewed to determine if this topic has already been explored, and these studies confirm the following statements about driving habits in the United States during and after the pandemic:

- Americans believe aggressive driving has increased since the COVID-19 pandemic, and super majorities believe that one or more of the following specific behaviors have increased: distracted driving, driving too fast, and reckless driving including weaving, tailgating, and running red lights. (Leppert, 2024, November 12)
- Fatalities per miles driven in May 2020 increased significantly over the same month in 2019. (NSC, 2020, July 21)
- Americans who admit to engaging in risky driving behaviors increased the amount of time they spent driving while safer drivers reduced the amount of time they spent driving during the first few months of the pandemic. (Tefft et al., 2022)
- The average number of daily trips taken by car dropped significantly in April 2020 during the COVID-19 lockdown compared to the last six months of 2019. (Tefft et al., 2021)

While these studies confirm important trends and patterns, none of them show if aggressive driving actually increased during COVID or if it has since declined or stabilized. They also do not show if the rate of accidents caused by at least one aggressive driver has changed or if any type of aggression (speeding, distraction, recklessness) or distractedness has increased more than others. Data scientists can answer these questions using appropriate data and modeling techniques. Data for this project are from the *PennDOT Pennsylvania Crash Information Tool* (*Pennsylvania Crash Information Tool*, n.d.), which are public datasets about crashes in the Commonwealth of Pennsylvania from 2005 through 2024. The objectives and modeling techniques for this project are summarized below:

Objective 1

Determine whether there has been a statistically significant increase in aggressive driving since the COVID-19 pandemic to provide evidence-based conclusions for PA Data Discourse readers about driving risks within 12 weeks.

Several different modeling approaches were pursued—including logistic regression (Hosmer et al., 2013), random forest (Breiman, 2001), and gradient boosting (Chen & Guestrin, 2016)—to identify if a relationship exists between COVID-19 and aggressive driving. To make sure the results were reliable, the data were split into separate training and testing groups so the model could be evaluated on data it had not already seen, and during modeling, data were repeatedly divided in different ways (5-fold cross-validation) to confirm that the findings are consistent and not dependent on a single split. The predictive performance of COVID-19 proved to be limited in all models.

The logistic regression model was selected for further evaluation because its results make it easier to understand how COVID-19 is associated with aggressive driving, even if limited. This further evaluation involved examining the estimated effect of COVID-19 to understand how

influential COVID-19 is in predicting aggressive driving. Time-based robustness checks were performed to ensure the results held up across different periods of time and not affected by varying distant past trends in the data. Together, these steps helped ensure that the model is both understandable and rigorously evaluated.

Objective 2

Identify specific aggressive driving behaviors that increased during COVID-19.

Several models using Support Vector Machines (SVM) one-class classification algorithm (Schölkopf et al., 2001 and scikit-learn developers, n.d.) were developed to use COVID-19 to predict specific aggressive or distracted driving behaviors instead of modeling aggressive driving with one indicator that encompasses all behaviors. The specific behaviors included proceeding without clearance; running red lights; running stop signs; speeding; speeding related, which includes speeding, driving too fast for conditions, or driver fleeing the police; tailgating; cell phone use; and distracted driving. These models were re-run after the modeling phase with corrected parameter values. However, this resulted in very little or no change in model metrics, depending on the target. No models for this objective met the technical success criteria of F1=0.7 established for this study, with F1 being a single measure that summarizes the model's accuracy in predicting crashes that involve aggressive driving and the model's ability to find all crashes that involved aggressive driving. The SVM models pushed the limits of available time and computational resources, even with a reduced dataset using only records after March 1, 2015. Therefore, these models were not pursued further to optimize or to obtain additional measures. Further study using different methods is recommended and outlined in the *1.2 Insights and Recommendations* section below.

Objective 3

Identify co-factors for crashes involving aggressive driving, such as age, fatigue, impairment, license status, hit and run, geography, or cell phone use.

Logistic regression was performed to identify if the relationship between different groups and aggressive driving changed with COVID-19. To make sure the results were reliable, the data were split into separate training and testing groups so the model could be evaluated on data it had not already seen, and during modeling, data were repeatedly divided in different ways (5-fold cross-validation) to confirm that the findings are consistent and not dependent on a single split. The predictive performance of COVID-19 among different groups proved to be limited. The estimated effects of each group relative to COVID-19 were examined to understand how influential COVID-19 is in predicting aggressive driving within each group. Time-based robustness checks were performed to ensure the results held up across different periods of time and not affected by varying historical trends in the data. Together, these steps helped ensure that the model is both understandable and rigorously evaluated.

Section 0: Executive Summary Presentation

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Section 1: Organizational Interpretation of Results

1.1 Translating Technical Results to Organizational Impact

Performance interpretation

Objective 1: Determine whether there has been a statistically significant increase in aggressive driving since the COVID-19 pandemic to provide evidence-based conclusions for PA Data Discourse readers about driving risks within 12 weeks.

The model to predict aggressive driving from COVID-19 demonstrated no meaningful power to distinguish between crashes that involved aggressive driving and those that did not. The Receiver Operating Characteristic (ROC) curve reinforces this conclusion. The ROC is a line or curve created by two rates plotted against one another: the rate at which the model correctly classifies crashes involving aggressive driving (true positive rate) and the rate at which the model incorrectly classifies crashes as involving aggressive driving (false positive rate). The ROC curve for this model, displayed in Figure 1 below, is perfectly aligned with the No Skill line, and the area under the curve (AUC), 0.5, means the model distinguishes between crashes that involved aggressive driving and those that did not about as well as random guessing. (Fawcett, 2006)

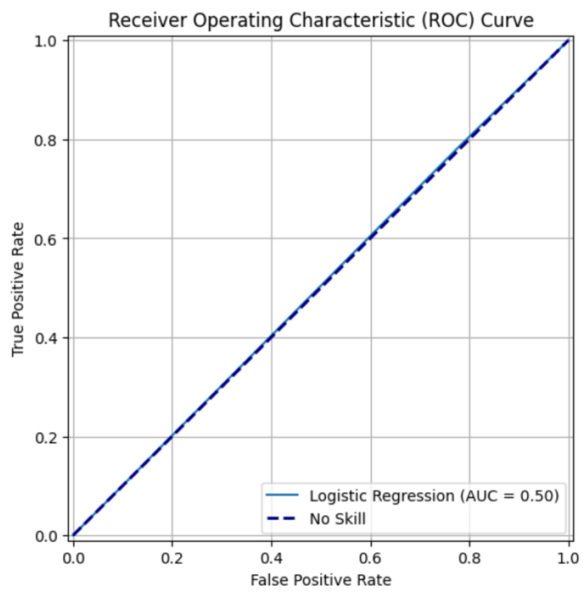


Figure 1 Receiver Operating Characteristic curve for objective 1

The confusion matrix as shown in Figure 2 below further illustrates the poor performance of the model. The model classifies 100% of crash records that involve aggressive driving as positive aggressive driving cases. However, the model misclassifies 100% of crashes that did **not** involve aggressive driving as positive aggressive driving cases. This confirms that the model cannot distinguish between crashes that involve aggressive driving and those that do not.

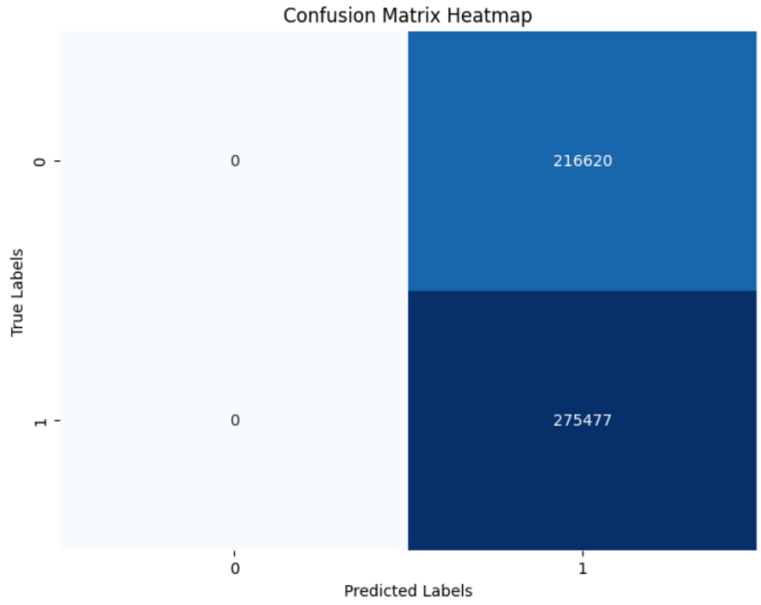


Figure 2 Confusion matrix for objective 1

Objective 3: Identify co-factors for crashes involving aggressive driving, such as age, fatigue, impairment, license status, hit and run, geography, or cell phone use.

The baseline model for this objective included indicators for post COVID-19, cell phone use, impaired driving, mature drivers, young drivers, being fatigued or asleep, hit and run drivers, unlicensed drivers, and urban or rural location. From this model, the four most important features — being fatigued or asleep, impaired drivers, young drivers, and cell phone use—were selected for a second model to determine if aggressive driving differs within each group before and during/after COVID-19. For example, this model could answer the question, did young drivers engage in more aggressive driving behaviors during and after COVID-19 than before?

The ROC curve for this model, displayed in Figure 3 below, is better than random guessing with an AUC of 0.58, however, this still fails to meet the technical success criteria established for this study, which is 0.7 or above. (Fawcett, 2006)

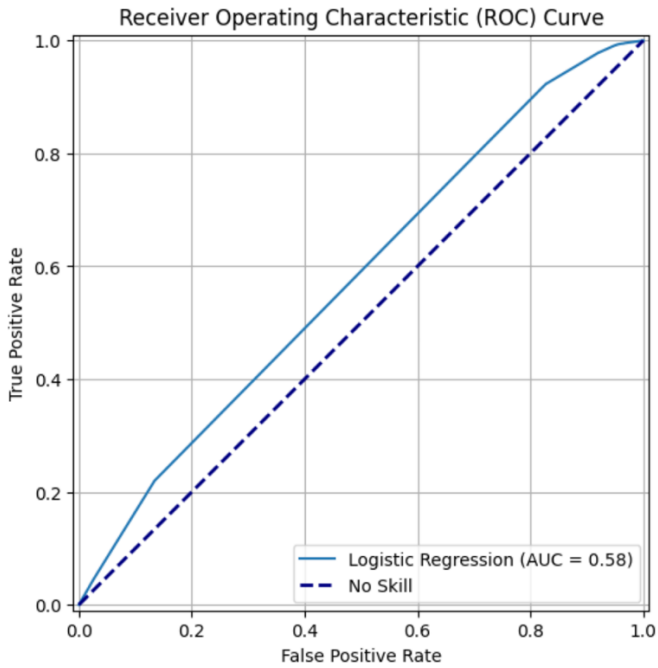


Figure 3 Receiver Operating Characteristic curve for objective 3

The confusion matrix as shown in Figure 4 below illustrates the modest performance of the model. The model correctly classifies only 59% of crashes—it correctly classifies 92% of crashes involving aggressive driving but only 17% of crashes that did **not** involve aggressive driving.

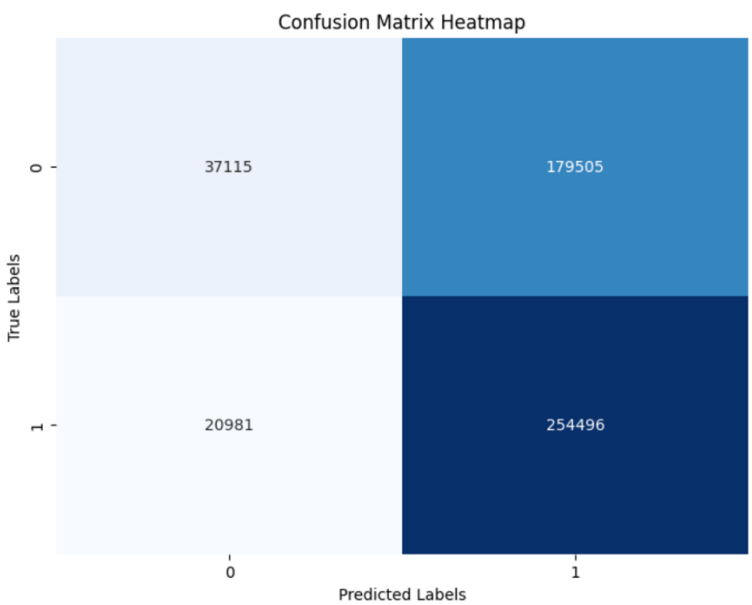


Figure 4 Confusion matrix for objective 3

Expected organizational outcomes

Publication of the results to inform the public is not recommended at this time. The model shows that there is no evidence to support the public's sentiment that aggressive driving increased

during COVID-19 and remained high. However, the dataset and methods used to reach this conclusion are just a select few in the large pool of possible data and methods. It is not appropriate or correct to conclude that aggressive driving did not increase during COVID-19 or remain high using only crash data. Not all aggressive driving leads to crashes. Different data and/or different methods may reveal that public sentiment is correct.

Confidence levels

ChatGPT. (n.d.) was referenced for this section. See *AI Statement of Use* below for specifics.

Objective 1: Determine whether there has been a statistically significant increase in aggressive driving since the COVID-19 pandemic to provide evidence-based conclusions for PA Data Discourse readers about driving risks within 12 weeks.

The model's ability to distinguish between crashes involving aggressive driving and those that do not is effectively no better than random guessing. This is indicated by the ROC AUC value of 0.503 (on a scale where 0.5 represents random chance and 1.0 represents perfect prediction). The 95% confidence interval for this estimate is [0.502, 0.504], meaning this narrow interval very likely contains the true value. The narrowness of this interval suggests the estimate is very precise, and the fact that it is centered around 0.5 reinforces the conclusion that the model has no meaningful predictive power. Figure 5 below shows the estimated mean ROC AUC with the 95% confidence interval, which is visually indistinguishable due to its narrow width.

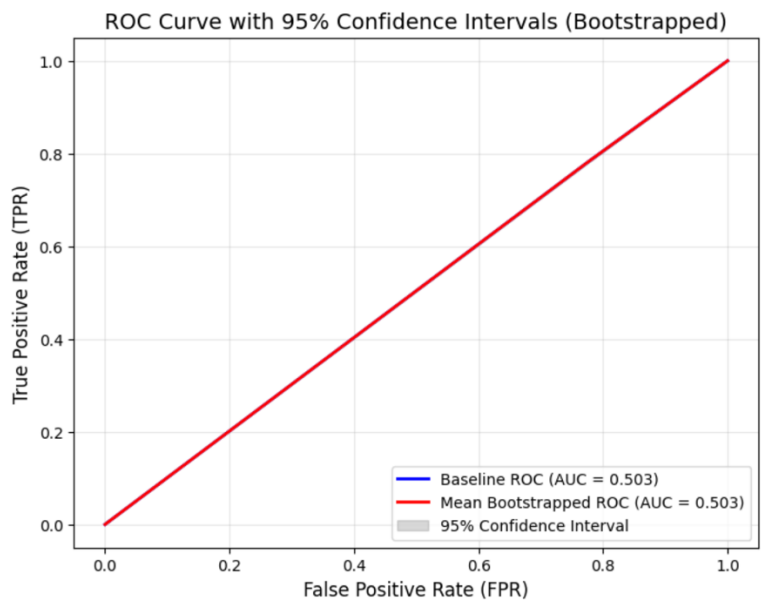


Figure 5 ROC curve with 95% confidence intervals for objective 1

To ensure these findings were not driven by distant past trends, two additional robustness checks were conducted using different subsets of the data. The first subset included one year before COVID-19 and the year 2020. The second included five years before COVID-19, the year 2020, and four years after. Across all three analyses—the original model and the two subsets—the estimated effect of COVID-19 on aggressive driving remained small and negative: $\beta \approx -0.035$

(95% CI [-0.041, -0.028]) in the original model, β ≈ -0.049 (95% CI [-0.067, -0.030]) in the first subset, and β ≈ -0.026 (95% CI [-0.034, -0.017]) in the second subset.

These coefficients indicate a slight decrease in aggressive driving associated with the COVID-19 period, but the effect is very small in magnitude. The consistency of these results across different time frames suggests that the findings are stable and not driven by how the data were segmented.

Overall, both the ROC AUC results and the robustness checks lead to the same conclusion: the model cannot meaningfully distinguish crashes involving aggressive driving based on COVID-19, and COVID-19 itself appears to have had little to no practical impact on aggressive driving behavior. There is no evidence in the crash data to suggest that aggressive driving increased during COVID-19 or remained high after.

Objective 3: Identify co-factors for crashes involving aggressive driving, such as age, fatigue, impairment, license status, hit and run, geography, or cell phone use.

The ability of the model to distinguish between crashes that involved aggressive driving and those that did not is limited. This is reflected in the ROC AUC value of 0.578, where 0.5 represents random chance and 1.0 represents perfect prediction. While this is slightly better than random guessing, the predictive performance is still considered weak. The 95% confidence interval [0.577, 0.580] is extremely narrow, indicating that this estimate is very precise and supporting the conclusion that the overall predictive ability is modest. Figure 6 displays the ROC curve and its confidence interval, which appears visually indistinguishable due to its tight range.

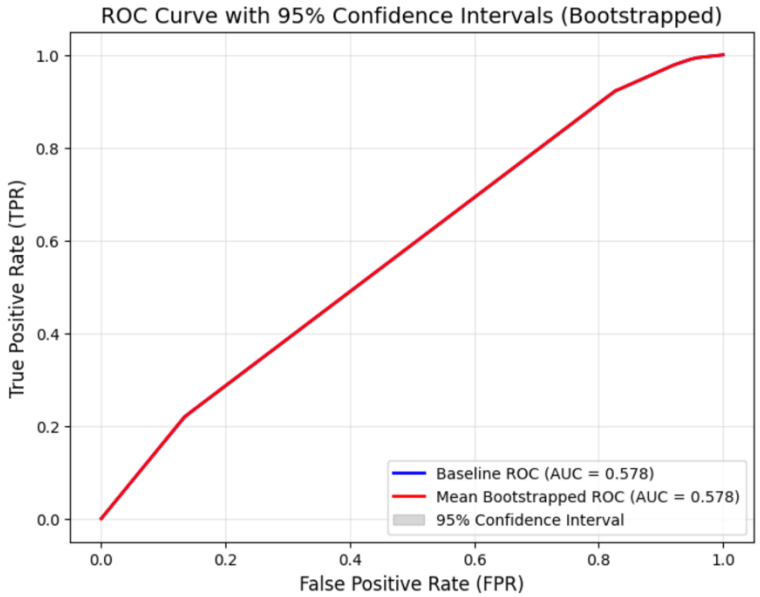


Figure 6 ROC curve with 95% confidence intervals for objective 3

Figure 7 presents the relative importance of model features, including being fatigued or asleep, cell phone use, impairment, young driver status, COVID-19, and the interaction between these variables and COVID-19. Among these, being fatigues or asleep has the strongest and most statistically significant effect (β ≈ -2.078, 95% CI [-2.110, -2.046]). This indicates a strong

association between being fatigued or asleep and aggressive driving. In contrast, the interaction between being fatigued or asleep and COVID-19 is not statistically significant ($\beta \approx 0.011$, 95% CI [-0.056, 0.077]), meaning there is no evidence that the relationship between fatigue-related crashes and aggressive driving changed during the COVID-19 period.

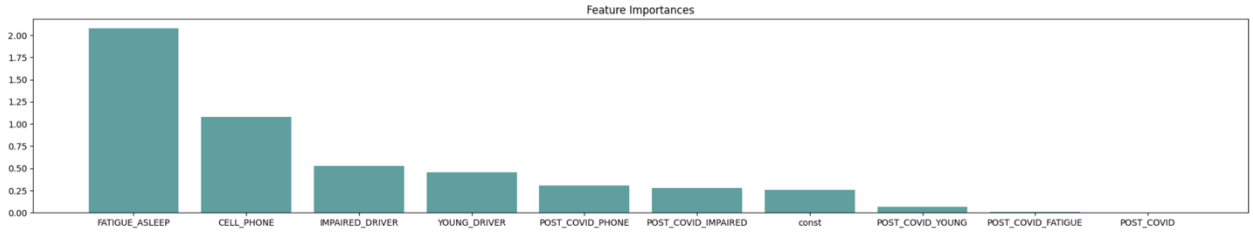


Figure 7 Feature importance for objective 3 final model with interaction

To assess whether these findings depend on distant past trends, two robustness checks were conducted using different subsets of the data. The first included one year before COVID-19 and the year 2020, and the second included five years before, the year 2020, and four years after. Across all three analyses—the original model and both subsets—being fatigued or asleep remained the most influential feature with no meaningful difference between pre-COVID and post-COVID periods.

Specifically, the interaction effect between fatigue and COVID-19 remained small and not statistically significant across all models: $\beta \approx -0.011$ (95% CI [-0.056, 0.077]) in the original model, $\beta \approx -0.042$ (95% CI [-0.141, 0.226]) in the first subset, and $\beta \approx -0.055$ (95% CI [-0.134, 0.023]) in the second subset. These results indicate that the role of being fatigued or asleep in crashes did not meaningfully change during COVID-19.

Overall, the results suggest that fatigue is the primary factor contributing to the model's (modest) predictive ability, and that fatigue-related crashes occur at similar rates regardless of whether they happened before, during, or after the COVID-19 period.

Value proposition

None of the models confirmed public sentiment that aggressive driving behavior increased during COVID-19 and remained high after. Additional analysis with different data and/or methods would require approximately two additional weeks of work over what was already planned for this project. However, a more thorough analysis would establish the publication's credibility in data-driven reporting and demonstrate its commitment to comprehensive analysis and public service reporting.

1.2 Insights and Recommendations

Key findings

As modeled, the data do not suggest an increase in aggressive or distracted driving during COVID-19 or after. However, some interesting findings occurred during exploratory data analysis that should be explored further because they support public sentiment about aggressive and distracted driving. Formatting the data as a time series and decomposing the data into the trend, seasonality, and random variation shows two important findings:

- Aggressive driving dropped during the COVID-19 lockdown but increased after.
- Some specific aggressive driving behaviors increased during COVID-19 and remained high after.

Key finding with aggressive driving

The trend in the decomposed time series in Figure 8 below shows that aggressive driving dropped during COVID-19 and then started increasing. Even though crashes involving aggressive driving have not increased beyond what they were before COVID-19, public sentiment may be based on the low point during the lockdown. In this sense, public sentiment may be correct.

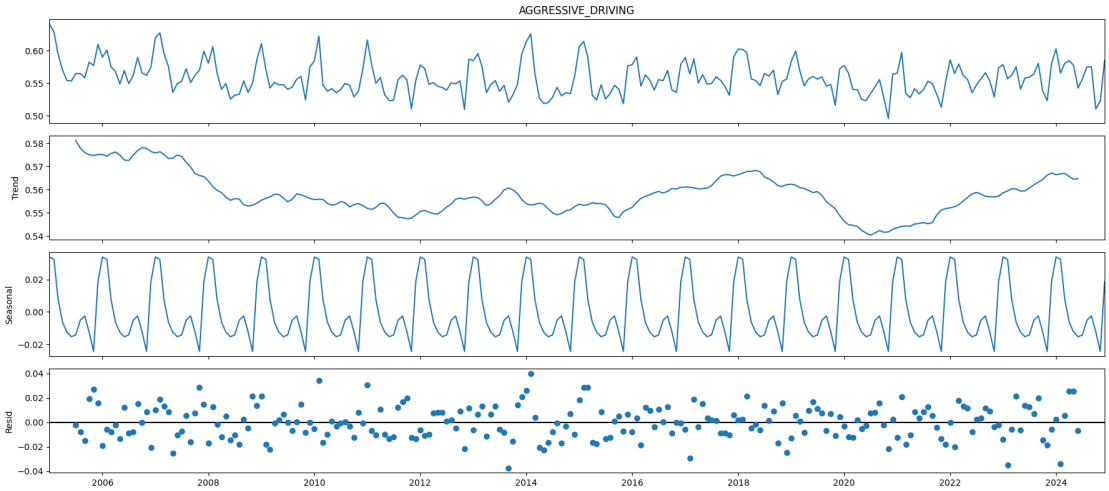


Figure 8 Time series decomposition for aggressive driving (AGGRESSIVE_DRIVING)

Key finding with specific aggressive or distracted driving behaviors

The trends in the decomposed time series plots in Figure 9 and Figure 10 suggest that proceeding without clearance and running red lights increased during COVID-19 and remain high. Figure 11 shows that cell phone use increased in 2022 and remains high. This suggests that public sentiment about aggressive and distracted driving may be correct.

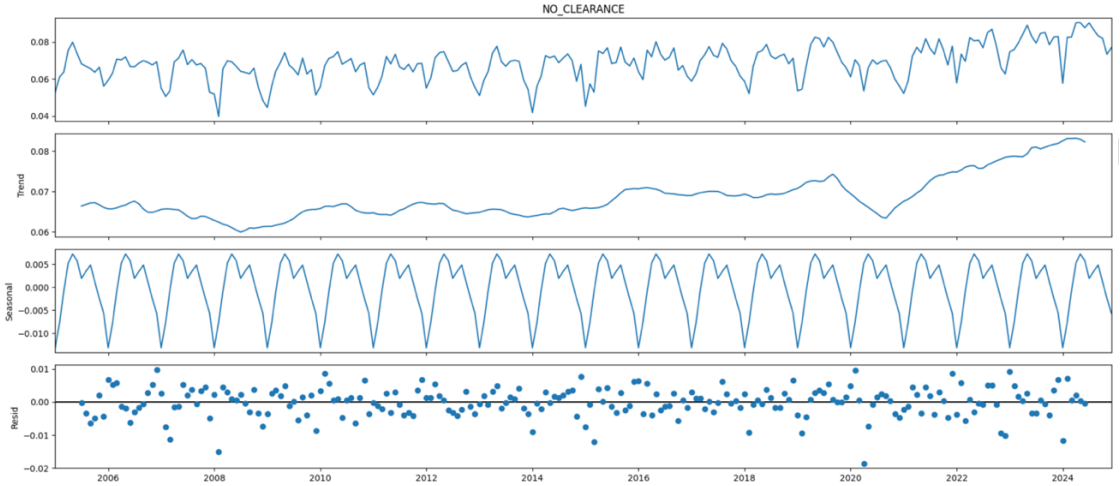


Figure 9: Time series decomposition for proceeding without clearance (NO_CLEARANCE)

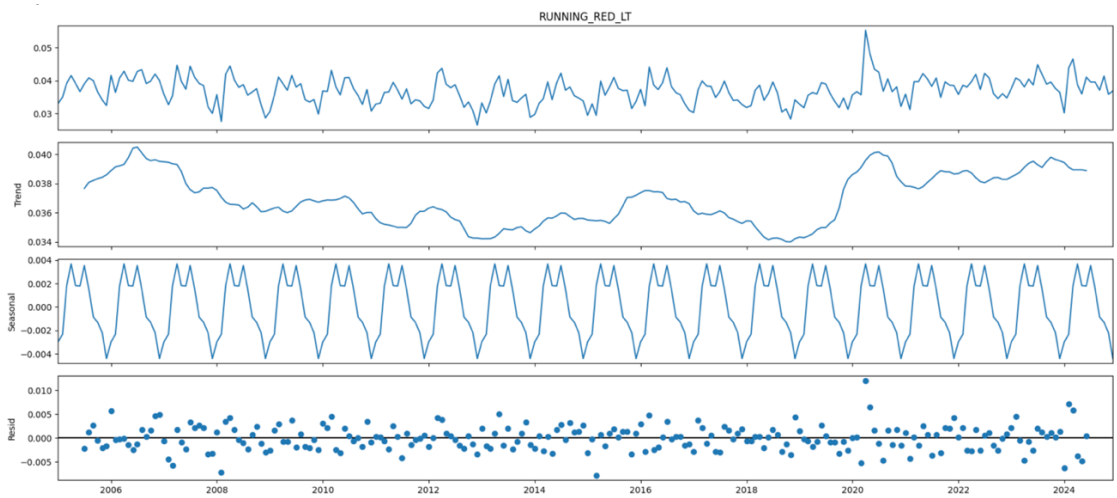


Figure 10: Time series decomposition for running red lights (RUNNING_RED_LT)

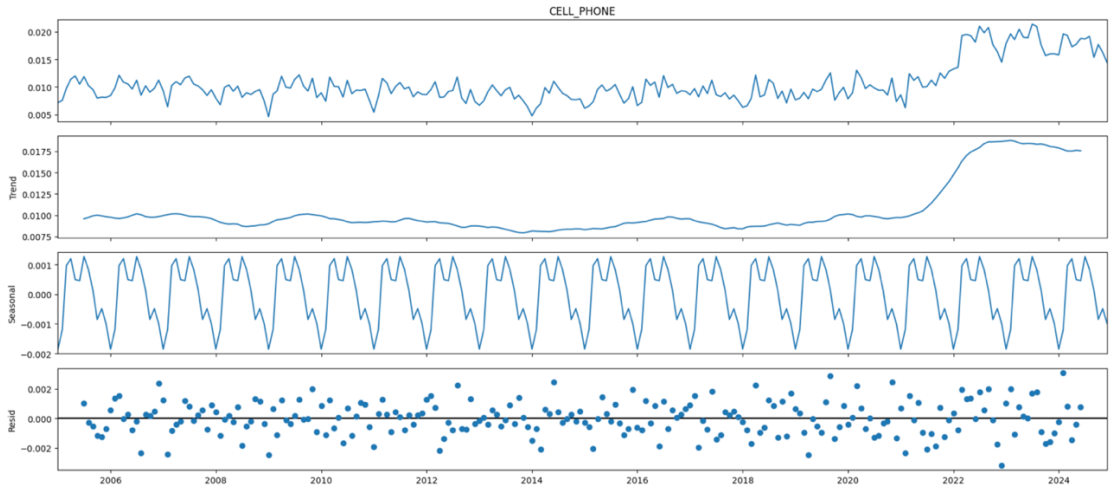
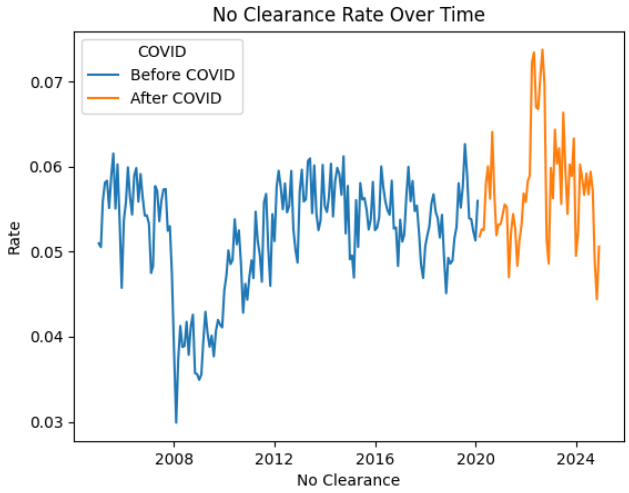


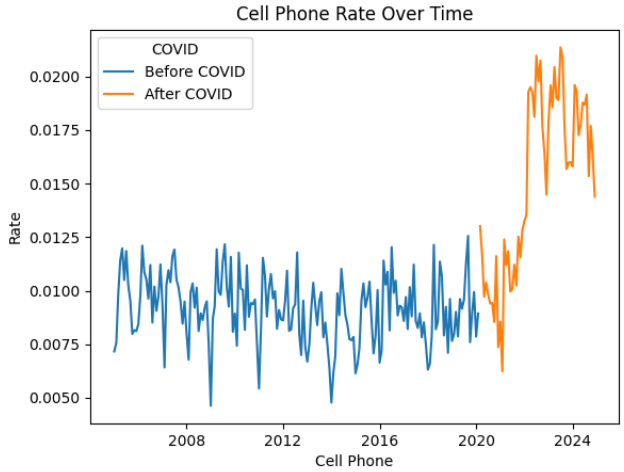
Figure 11: Time series decomposition for cell phone use (CELL_PHONE)

Surprising discoveries

The fact that the F1 score was very low or 0.0 for the individual driving behaviors modeled with one-class classification is surprising given some of the time series plots, which were developed to provide accessible visualizations for public understanding. Two examples are shown in Figure 12 below.



Proceeding without clearance increased slightly and then dipped during COVID-19 in 2020. The behavior then spiked significantly in 2023 before trending down.



Cell phone use spiked after COVID-19 in 2022 or 2023.

Figure 12 Proceeding without clearance and cell phone use changes after COVID-19

Although neither of these behaviors spiked at the start of COVID-19, it is possible that these spikes were still related to COVID-19. It makes sense that crashes related to vehicles proceeding without clearance or cell phone use would decrease, or at least not increase, during COVID-19 when there were fewer vehicles on the road. When people started driving again, it makes sense that they continued the habits they started during COVID-19, and with more cars on the road, this contributed to more accidents. Proceeding without clearance is trending down, but it was still higher than it was before COVID-19 for a couple of years.

Strategic recommendations

Several possibilities exist for moving forward with additional research given the poor performance of the classification models and the patterns revealed by the time series plots.

Recommendation	Rationale
Model the time series with ARIMA	The decomposition of specific aggressive and distracted driving behaviors shows trends that increased during COVID-19 and remained high. Modeling the pre-COVID data with ARIMA may show that the predicted trends diverge from the post-COVID trends, which would indicate change points in the data.
Acquire additional data	Crash data may not be the best source for identifying if aggressive and/or distracted driving increased during COVID-19 and remained high. Not all aggressive or distracted driving behaviors result in crashes. Drivers will notice when other drivers are aggressive or distracted even when they are not involved in or witnesses to a crash. Police traffic citation data are public and could be acquired for additional analysis.

Feature importance interpretation

Objective 1: Determine whether there has been a statistically significant increase in aggressive driving since the COVID-19 pandemic to provide evidence-based conclusions for PA Data Discourse readers about driving risks within 12 weeks.

The estimated effect of the COVID-19 indicator on aggressive driving is negative but very small in magnitude ($\beta \approx -0.035$), and the 95% confidence interval ([-0.041, -0.028]) confirms that COVID-19 has an effect and that the effect is very small—too small to be practically important. The small effect size indicates that COVID-19 is no more than mildly associated with aggressive driving.

Objective 3: Identify co-factors for crashes involving aggressive driving, such as age, fatigue, impairment, license status, hit and run, geography, or cell phone use.

Figure 13 below illustrates the effects of fatigue, cell phone use, impairment, young driver, COVID-19, and the association between these factors with COVID-19.

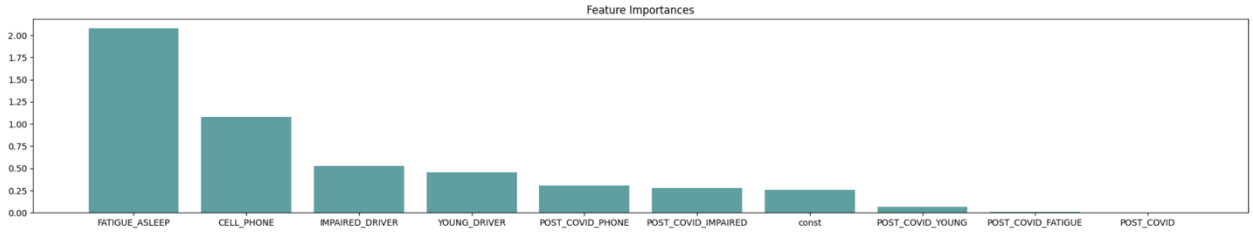


Figure 13 Feature importance for objective 3 final model with interaction

The estimated effect of fatigue ($\beta \approx -2.078$) with 95% confidence interval [-2.110, -2.046] suggests that being fatigued or asleep is the main effect responsible for the modest ROC AUC. The estimated effect of fatigue and COVID-19 ($\beta \approx 0.011$) is not statistically significant. The

95% confidence interval [-0.056, 0.077] includes zero, which suggests that there may be no association between being fatigued or asleep and COVID-19. Crashes that involved being fatigued or asleep are consistent regardless of when they occurred relative to COVID-19.

The estimated effects of the associations between COVID-19 and each of the other three features—cell phone use, impaired driving, and young driver—are statistically significant but small. The effect of cell phone use and COVID-19 on aggressive driving ($\beta \approx -0.307$) is confirmed to be modest with 95% confidence interval [-0.375, -0.238]. The effect of impaired driving and COVID-19 ($\beta \approx 0.281$) is confirmed to be modest with 95% confidence interval [-0.305, -0.257]. The effect of young drivers and COVID-19 ($\beta \approx 0.066$) is confirmed to be negligible with 95% confidence interval [0.048, 0.085]. This suggests that, relative to COVID-19, cell phone use, impaired driving, and young drivers are each only mildly associated with aggressive driving and of negligible practical importance.

Limitations and caveats

These models cannot identify patterns of aggressive driving that did not result in a crash. If police citation data are collected and analyzed, those models would identify more instances of aggressive driving, however, not all aggressive driving results in a citation, either. Care must be taken when reporting results to make sure these limitations are understood.

1.2 Risk Assessment and Mitigation Strategies

Publishing the results of these models and stating that public sentiment is not supported by the data would be premature. If incorrect, PA Data Discourse risks losing the attention and trust of readers. The estimated time for this additional research is only two weeks, and the publication would be in a better position to provide an accurate and meaningful answer to the public.

Model risks

As a historical study, there are no risks with modeling aggressive driving using COVID-19 as a predictor. The results from the models, including any additional models pursued as recommended, are focused on answering a question about events prior to and after one particular point in time.

If PA Data Discourse pursues ongoing monitoring of aggressive driving through a public dashboard, the objectives would change from answering questions about the effect of COVID-19 on aggressive driving to objectives centered around other influential effects or to simply monitoring the trends. Model risks would have to be evaluated with respect to these new objectives.

Data risks

Availability of data is the number one data risk with this project. The current models are a historical view of the data, however, if additional data are pursued for further study or ongoing monitoring, there are risks that some or all of the desired data will be unavailable either now or at some point in the future.

Organizational risks

If PA Data Discourse publishes the current models, readers may realize that the research was not comprehensive and become dissatisfied. The reputation of the publication as a public service organization would be damaged. Even if further research still shows no evidence to support

public sentiment that aggressive or distracted driving increased during COVID-19 and remained high, readers will notice and appreciate a more comprehensive review of events.

Ethical considerations

There are no ethical considerations with this study. The crash datasets contain data that may enable people to trace crashes and identify people, however, PA Data Discourse has not used these variables. The demographic variables used in this study are aggregated counts. For example, the URBAN_RURAL flag simply indicates if a crash happened in an urban or rural area. The YOUNG_DRIVER and MATURE_DRIVER flags prevent specific ages from being known. Any data published from this study will only include the variables used, not the entire set of variables available from PennDOT.

Mitigation approaches

To mitigate the model and data risks, it is recommended that PA Data Discourse honor its commitment to publishing accurate, well-researched, public service content only when all possible and feasible research actions have been pursued. Conducting this additional research in a timely manner will honor the publication's commitment to provide current content of interest to the public.

When the research is published, PA Data Discourse will seek peer review for the article and will monitor the performance of the models deployed on the dashboard. Data journalists will also engage with the public via the post's comment section to address questions and to provide a venue for modifying or retracting published results.

Section 2: Project Review and Reflection

2.1 Project Retrospective

What went well

The fact that the ROC AUC measures from either model were not consistent with the corresponding F1 measures brought attention to the imbalance between precision and recall and allowed a better interpretation of the models, which strengthened the analysis. The time series decomposition plots provided support for the key findings and recommendation section, which includes a recommendation to model the pre-COVID data with ARIMA to see if predicted trends diverge from the post-COVID trends, indicating change points in the data.

Challenges faced

The models took a very long time to run, which caused problems with the laptop running too hot for too long. This was mitigated by using Google Colaboratory for the modeling. Even though the laptop was not stressed anymore, the models still took a long time to run, which cut into the time allotted for evaluation. Grid search may not be the best approach to finding the best parameters. Grid search was abandoned for the new modeling done for the final report because of the time required to run all of the iterations.

Alternative approaches

Alternatively, more time discussing the features with the Crash Analysis Manager for PennDOT would have been productive and may have led to the selection or creation of different features. Based on the time series plots, modeling with ARIMA first would have been a better choice.

Analyzing whether predicted trends from pre-COVID analysis diverged from the post-COVID actual trends would reveal if change points exist related to COVID-19.

CRISP-DM reflection

The CRISP-DM framework kept the project grounded in the primary and two secondary objectives. It would have been easy to include more features, but this would have distracted from the goals outlined in the business and data understanding phases. Following the framework kept me focused and provided a scaffold for documenting the project.

Time allocation

Spent 40% of the time exploring the context and the data for the business and data understanding phases. Spent 50% of the time modeling. This was too much time as it left only 10% of the time for evaluation. Guidance and a productive review of evaluation concepts helped bring the project together in the end.

2.2 Lessons Learned and Future Applications

Technical lessons

Many technical skills and better understanding were gained from this project. The project served as a good review of data analysis concepts including statistical significance testing and interpretation, visualizations that tell a story, setting up pipelines, and hyperparameter tuning. New skills were developed including feature importance visualizations and bootstrapping to obtain confidence intervals for the ROC AUC.

Process insights

Discovered that it was helpful to create a document with the CRISP-DM framework sections as headings to use as a template. This template was filled in with content as the project progressed when steps taken and their rationale were fresh. Discovered that the literature review and business context understanding completed at the beginning of the project helped to keep the project grounded and focused on the stated objectives.

Domain knowledge

Gained a deeper understanding of factors that do and do not contribute to crashes. Learned how to communicate technical performance measures in the business context of crashes and aggressive driving for a non-technical audience.

Tool proficiency

Already proficient in data exploration with Pandas. Relearned how to set up preprocessing to transform text columns into one-hot encoded columns and how to set up a pipeline to complete all preprocessing together. Learned how to conduct bootstrap resampling using scipy interpolate and the model's predicted probabilities to create confidence intervals. This involved sampling the test set multiple times, calculating the ROC curve and AUC for each set, and then estimating the mean ROC curve and AUC confidence intervals.

Future applications

The data processing and modeling techniques could be adapted for any dataset in any domain. The model evaluation framework developed in this project including the feature importance

visualizations and bootstrapping to create confidence intervals for performance measures could be adapted to any modeling task.

Conclusion

This project examined whether aggressive driving in Pennsylvania changed during and after the COVID-19 pandemic, using statewide crash data from 2005 through 2024. The key result is straightforward: COVID-19 does not meaningfully predict aggressive driving in crash data. The model's performance was no better than random guessing, meaning it cannot reliably distinguish between crashes that involved aggressive driving and those that did not. While there is a statistically detectable relationship, the size of that effect is extremely small and not practically meaningful.

Some additional patterns are worth noting. Time-based analysis shows that certain behaviors—like running red lights and cell phone use—did increase after the initial lockdown period and remained elevated for some time. This may help explain why public perception differs from what the overall model shows. Additional study is recommended to provide the public with as comprehensive of a review as possible. Modeling crash data with ARIMA may reveal that public sentiment about aggressive driving is correct.

AI Statement of Use

AI (*Chatbot App—AI Chatbot*) was used to troubleshoot errors, especially errors related to setting up the virtual environment. It was also used to look up code syntax and definitions of terms for better understanding. AI was not used for coding. One section of text (*Confidence levels* in *1.1 Translating Technical Results to Organizational Impact*), written in its entirety by me, was copied into ChatGPT for suggestions on rewording for a non-technical audience. The result was then edited significantly before pasting back into this report.

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